

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 50
- max_units: 150
- units_step: 50
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam']

Tuning:

- max_epochs: 100
- hyperband_iterations: 3

BiLSTM Multi-Step Training Report

Report Generated: 2025-05-02 16:59:11

Training Start: 2025-04-26_23-37-18

Training End: 2025-04-26_23-46-45

Training Duration: 9.45 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 0.9947

MSE: 1.6036

RMSE: 1.2663

R2 Score: 0.7349

Multi-Step Forecast:

Forecast Steps: 10

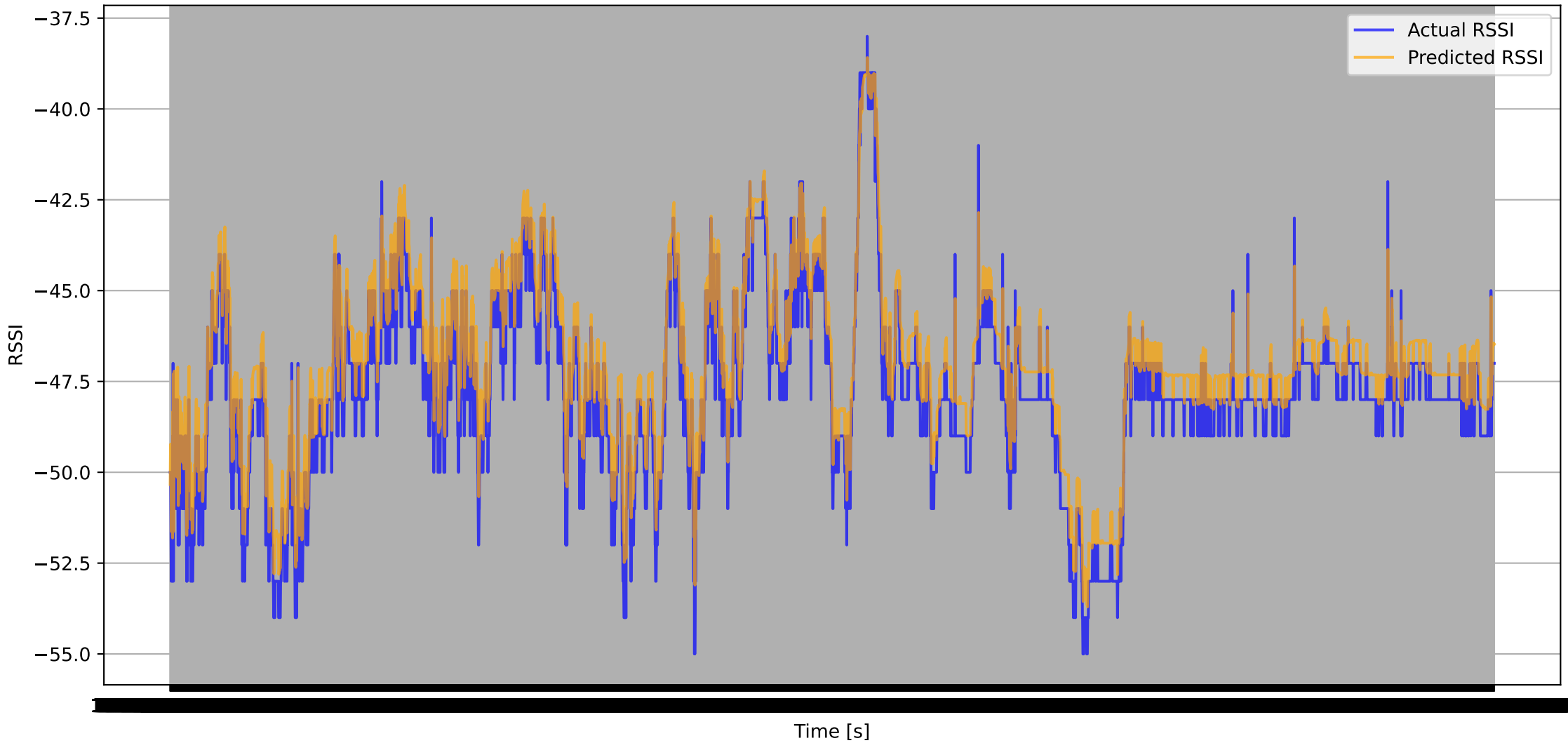
MAE: 0.0305

MSE: 0.0013

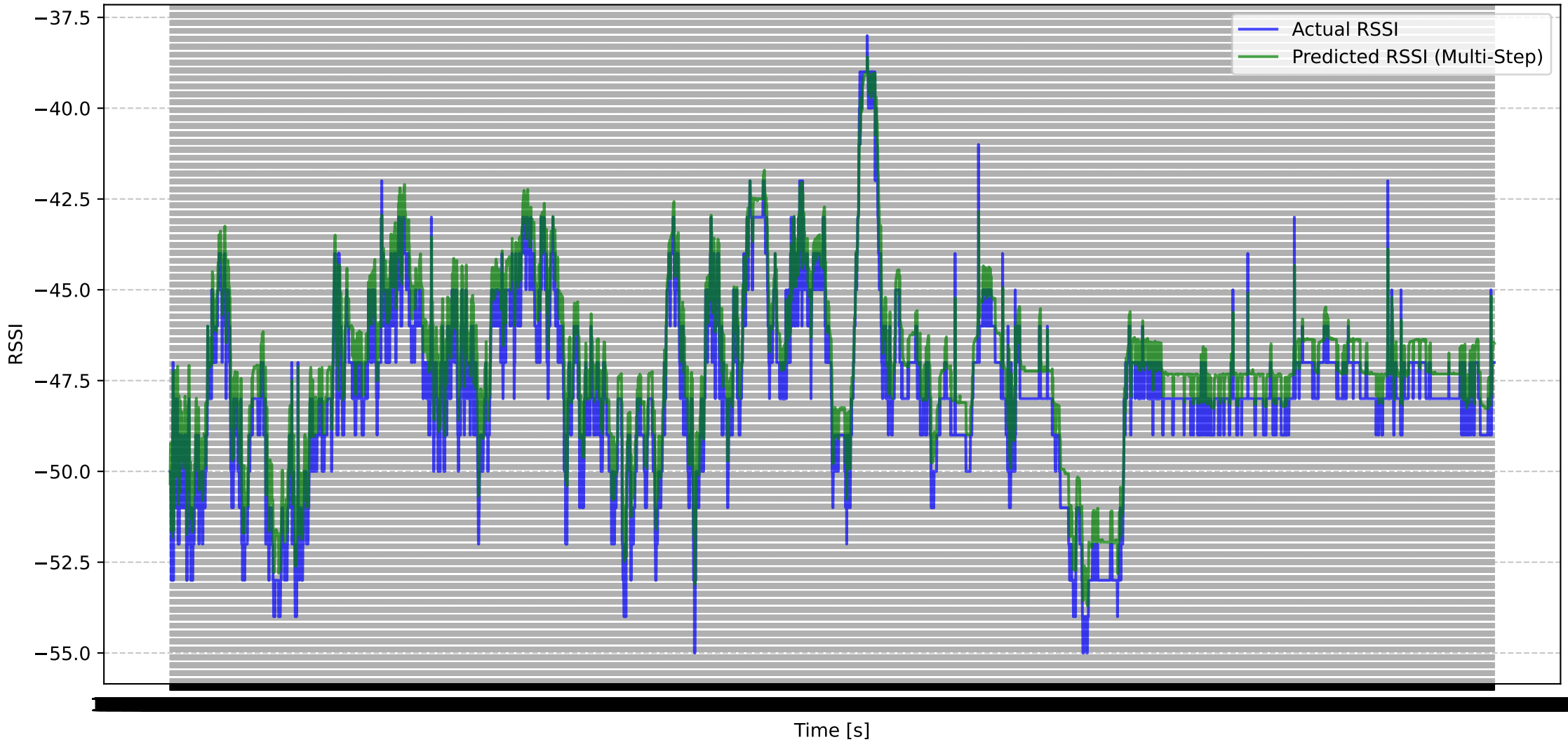
RMSE: 0.0354

R2 Score: 0.8265

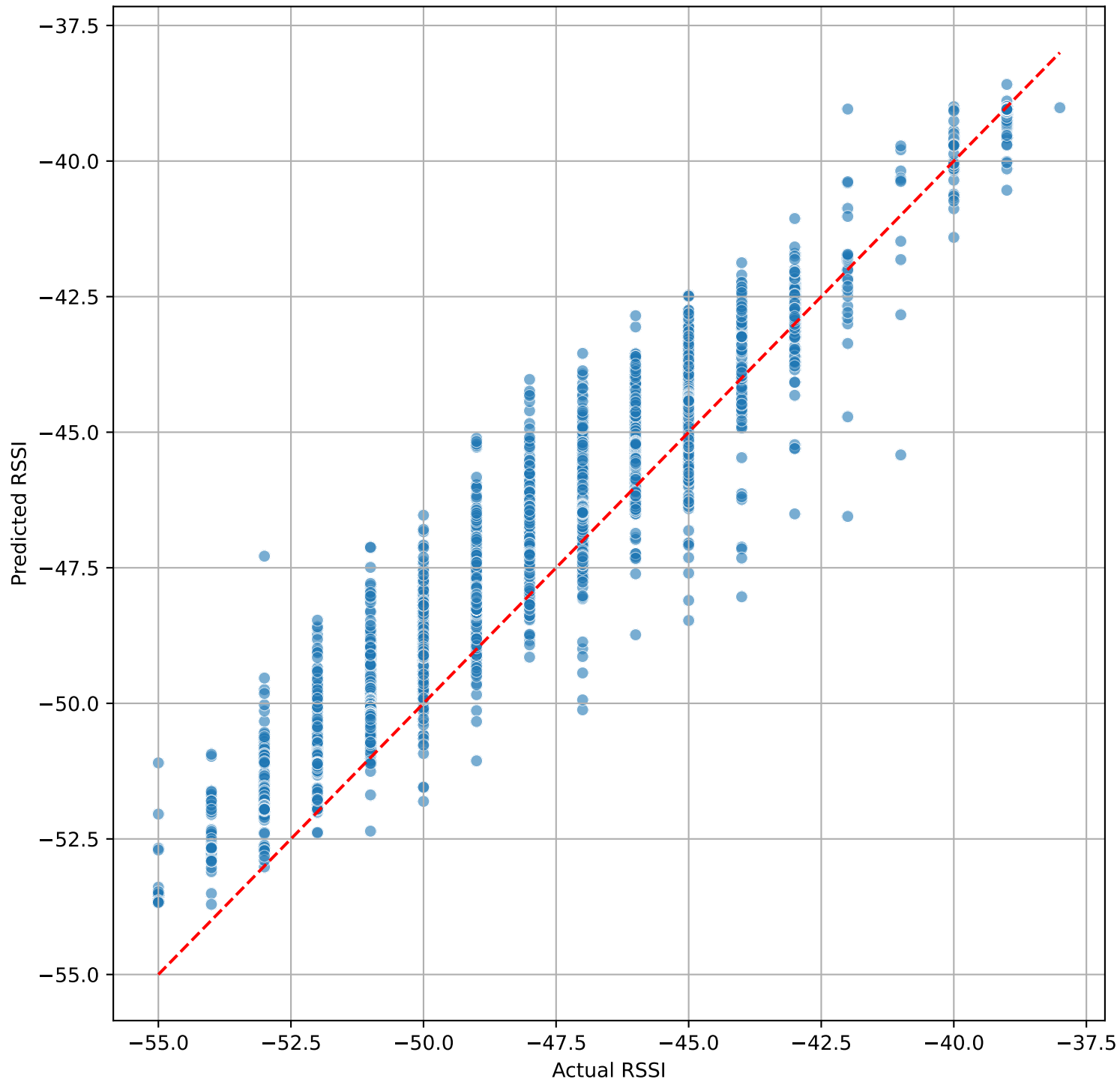
Actual vs Predicted RSSI (First Step)



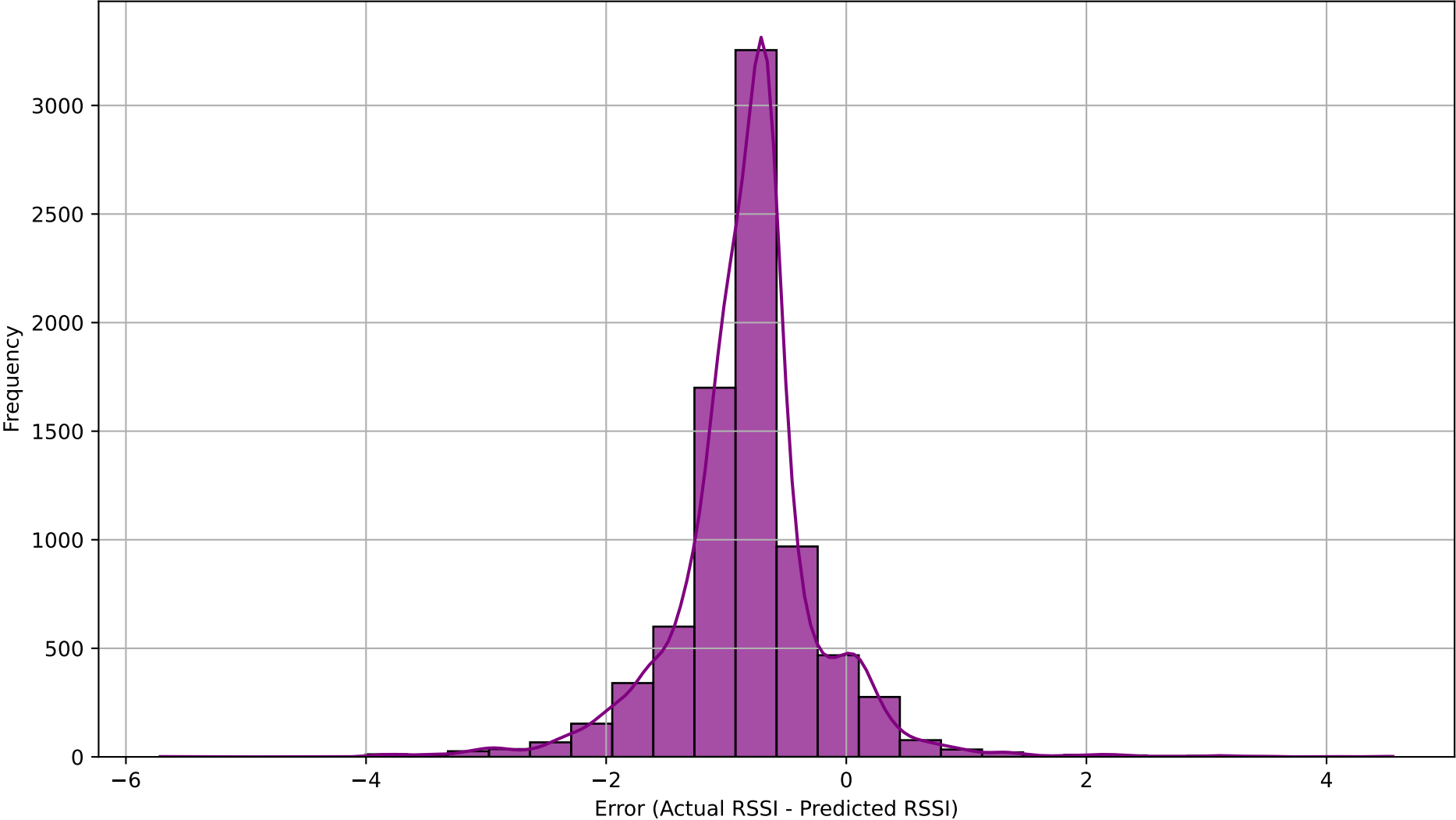
Actual vs Predicted RSSI (First Step vs Multi-Step)



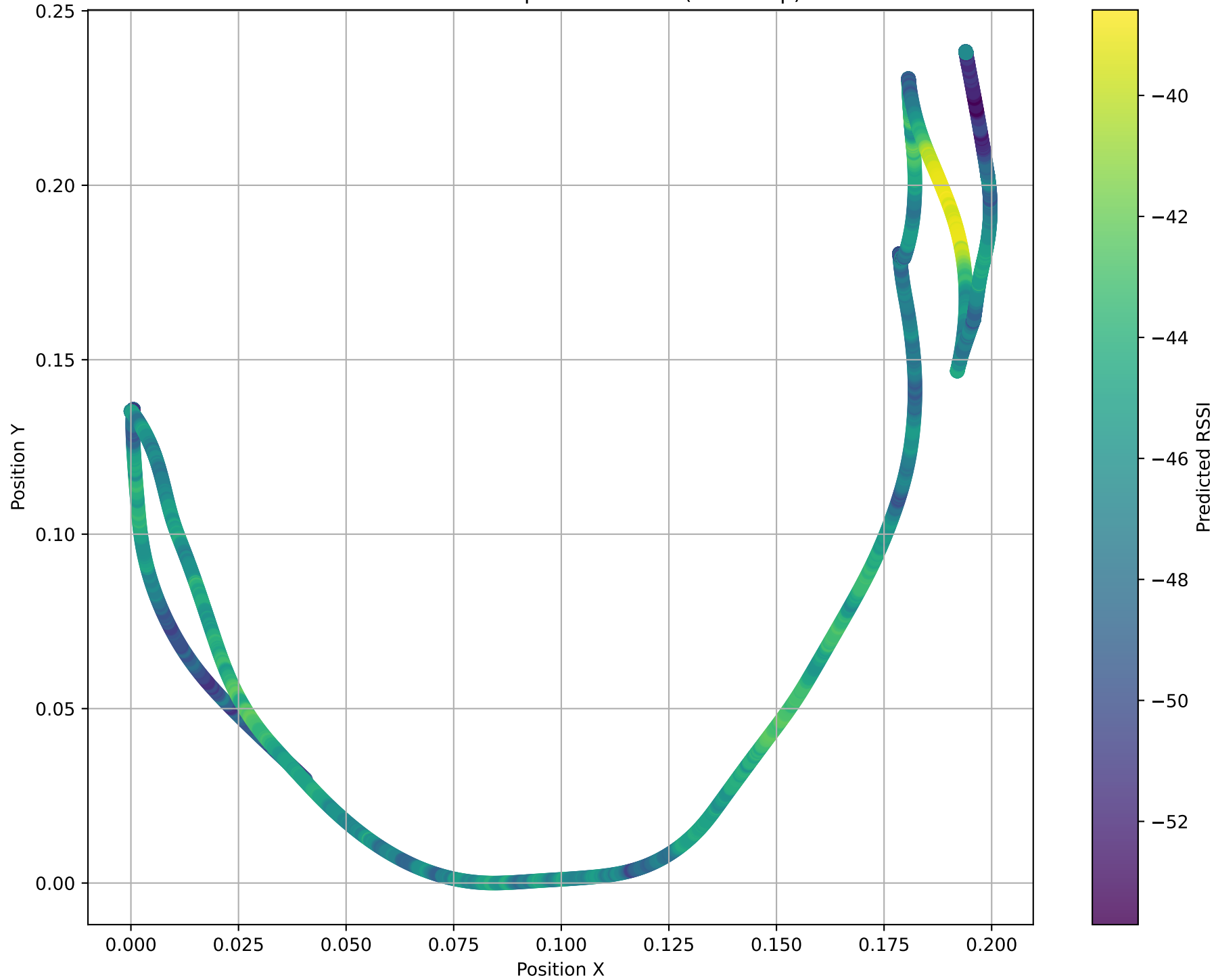
Actual vs Predicted RSSI (First Step)



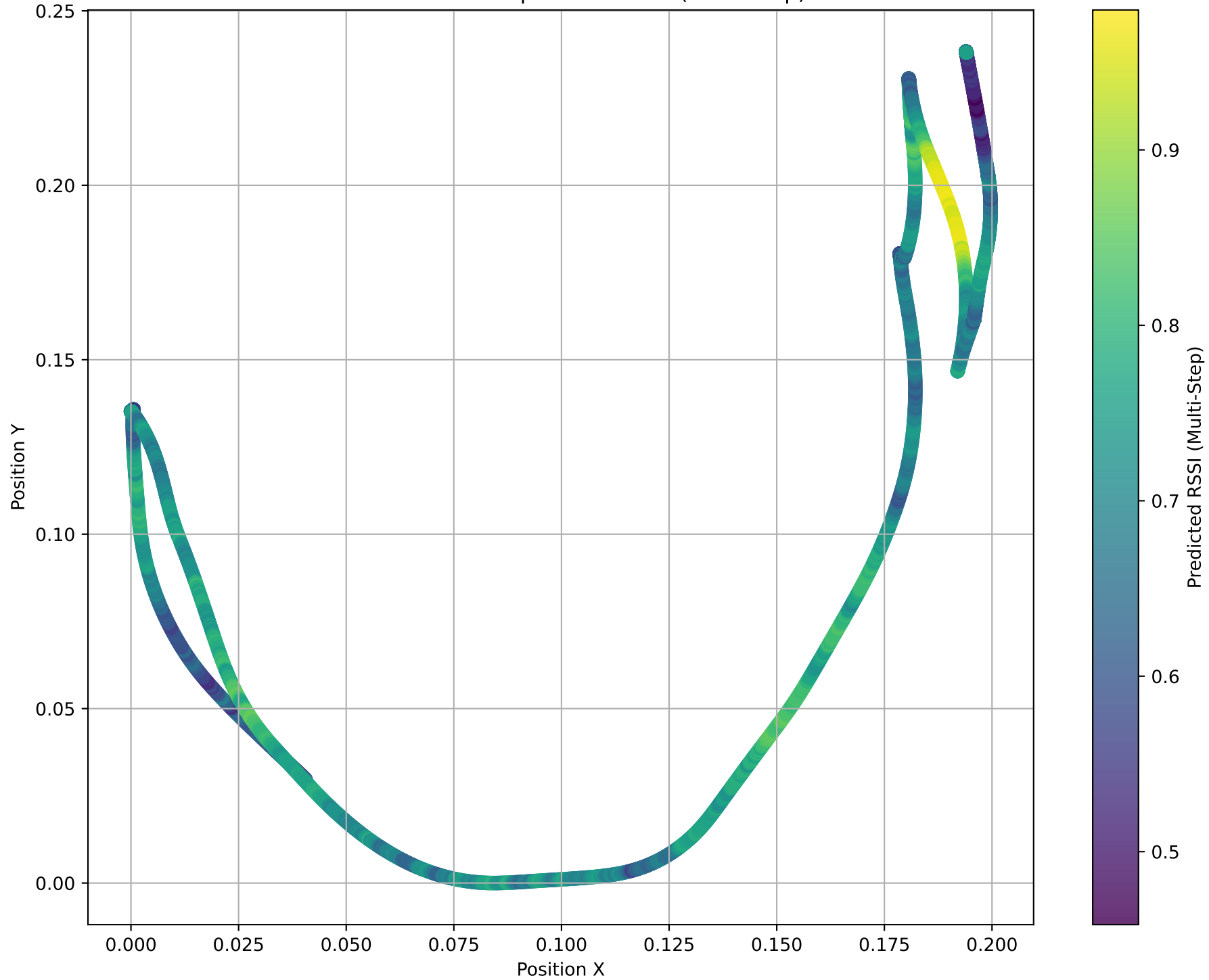
Distribution of Prediction Errors (First Step)



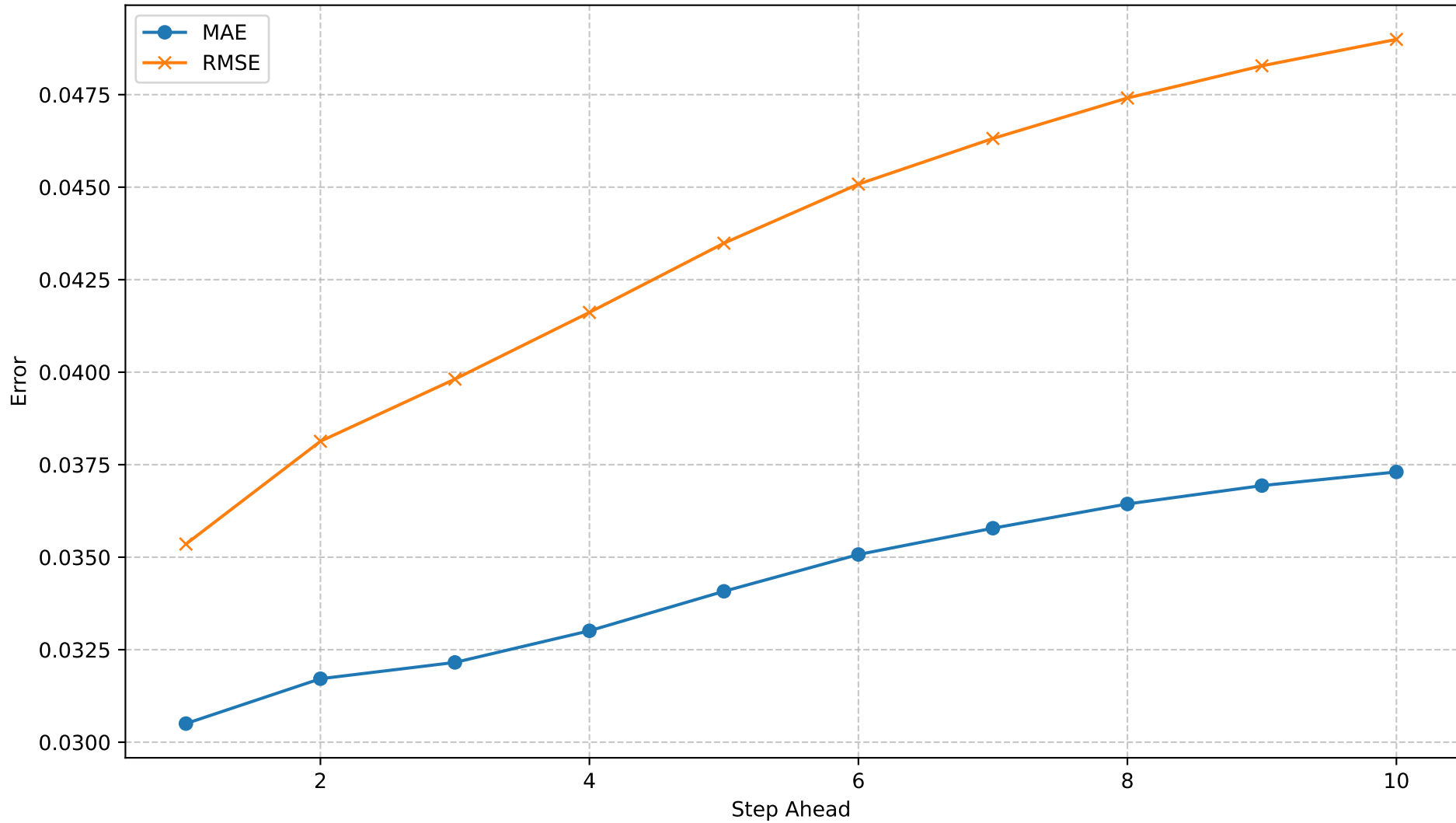
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step



Resumo Automático:

- Forecast Steps configurados: 10
- Passos com MAE ≤ 2.0 : 10 passo(s)
- MAE no último passo: 0.0373

[V] O modelo mantém boa acurácia durante quase toda a sequência de previsões.